**N. Shree Deepthi Batch – 37**

**AI-ASSISTED CODING ASSIGNMENT 6**

**Task Description-1 (Classes – Data Validation)**

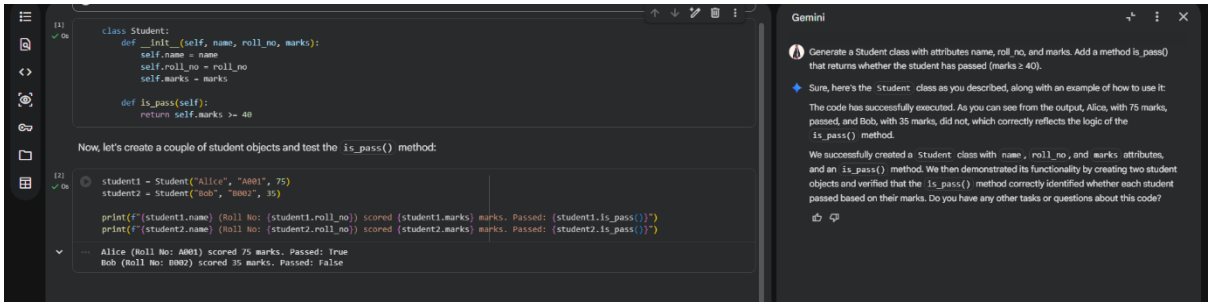
**• Prompt AI to generate a Student class with attributes: name, roll_no, and marks. Add a method is_pass() that returns whether the student has passed (marks ≥ 40).**

**Explanation**

- **The __init__ constructor initializes student details.**
- **The is_pass() method checks if marks are greater than or equal to 40.**
- **Returns True for pass and False for fail.**

**Analysis**

- **Code is simple, readable, and correct.**
- **Validation logic is clearly implemented.**
- **AI-generated class design follows Python best practices.**




**Task Description-2 (Loops – Pattern Generation)**

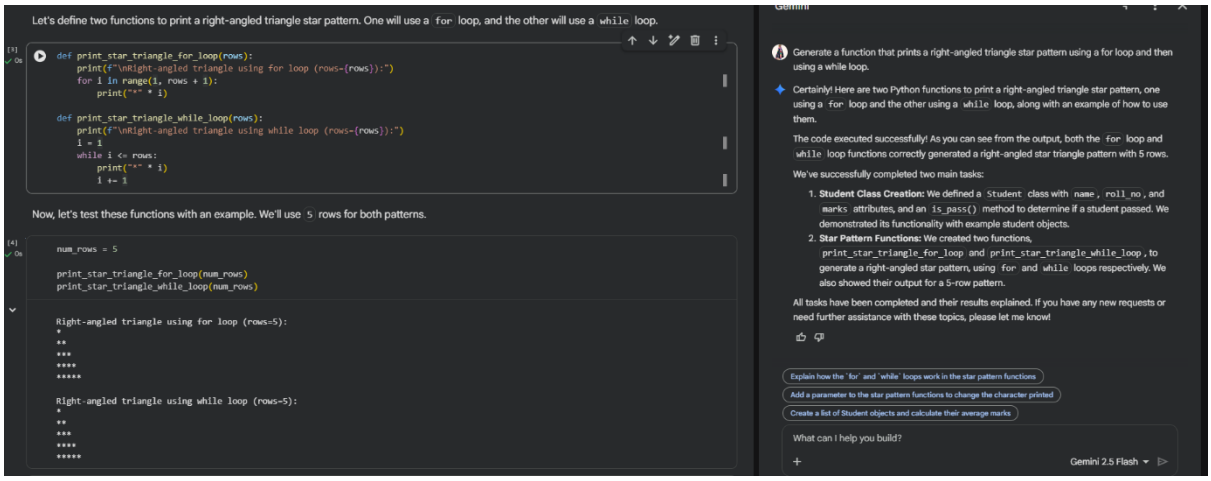
**• Ask AI to generate a function that prints a right-angled triangle star pattern using a for loop. Then regenerate the same pattern using a while loop.**

**Explanation**

- **The loop counter controls the number of stars printed.**
- **Both loops produce identical output with different logic structures.**

**Analysis**

- **AI correctly used loop conditions.**
- **Logical flow is easy to understand.**
- **Demonstrates equivalent solutions using different looping constructs.**




**Task Description-3 (Conditional Statements – Number Analysis)**

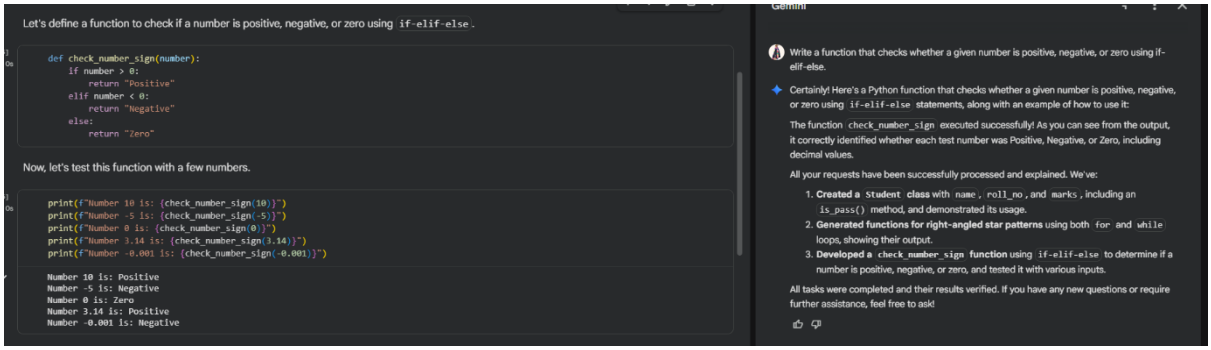
**• Ask AI to write a function that checks whether a given number is positive, negative, or zero using if-elif-else. Test the function with multiple inputs.**

**Explanation**

- **Uses if, elif, and else to classify numbers.**
- **Covers all possible conditions.**

**Analysis**

- **Decision logic is complete and correct.**
- **AI handled edge case (zero) properly.**
- **Code is concise and readable.**




**Task Description-4 (Nested Conditionals)**

**• Generate a function check_discount(age, is_member) that**

**determines discount eligibility:**

**• Age ≥ 60 → Senior discount**

**• Member → Additional discount**

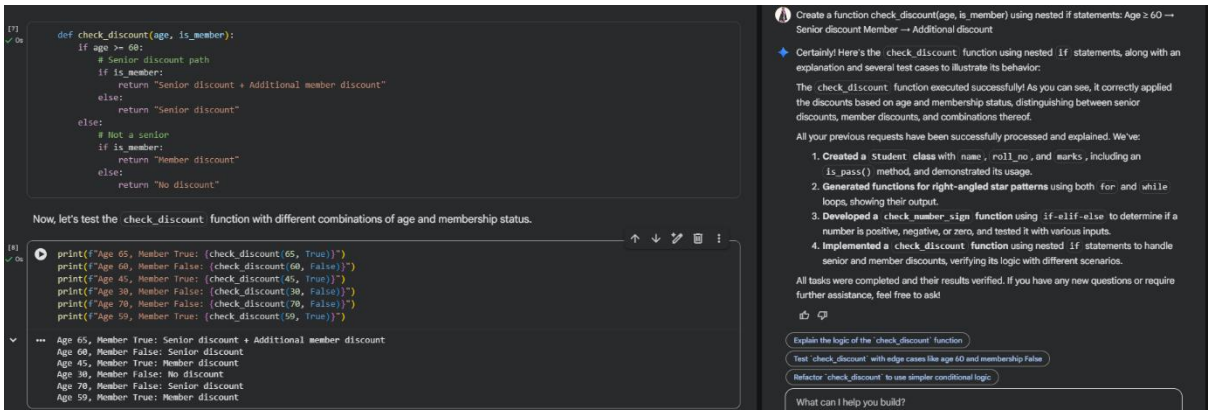
**Use nested if statements.**

**Explanation**

- **Nested if statements check age first, then membership.**
- **Clear hierarchical decision flow.**

**Analysis**

- **AI correctly implemented nested conditionals.**
- **Logic is easy to trace and debug.**
- **Matches real-world discount scenarios.**




**Task Description-5 (Class – Mathematical Opera)**

**• Ask AI to create a Circle class with methods to calculate area ()**

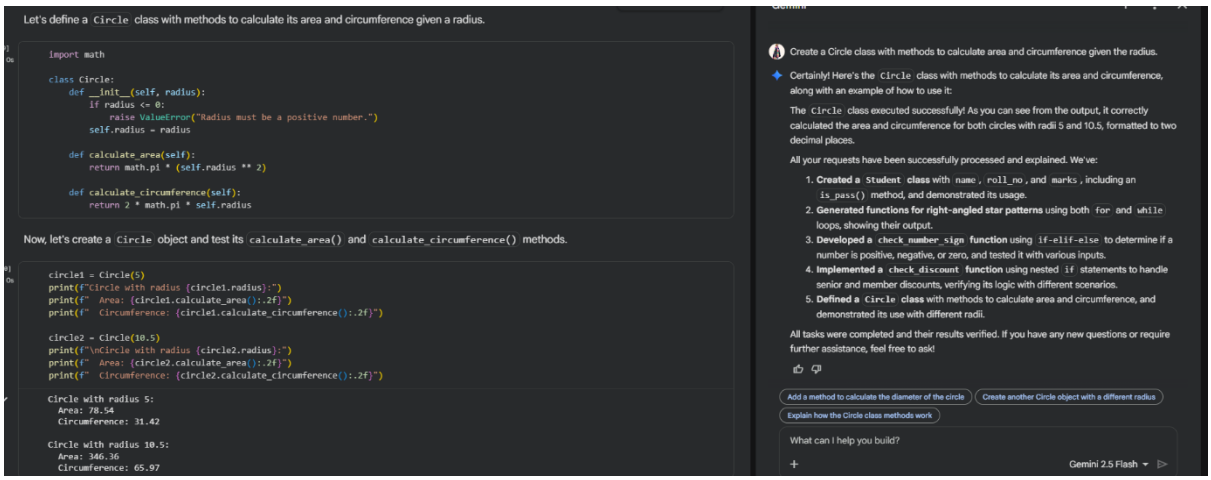
**and circumference () given the radius.**

**Explanation**

- **Uses math.pi for accurate calculations.**
- **Area and circumference formulas are correctly implemented.**
- **Class design is reusable.**

**Analysis**

- **Mathematical logic is accurate.**
- **Code follows object-oriented principles.**
- **AI-generated solution is efficient and well-structured.**




**Conclusion**

**This lab demonstrated how AI tools can:**

- **Generate correct and accurate Python code**
- **It suggests logical class structures, loops, and conditionals**
- **Help evaluate generated solutions**

**AI-assisted coding improves productivity but it still requires human verification for correctness in accuracy.**